GENOME-SCALE PERTURB-SEQ · ATOPIC DERMATITIS

Beyond STAT6

Which intracellular nodes collapse the type-2 program — and which are druggable?

Sayane Shome, PhD

linkedin.com/in/sayaneshome

github.com/sayaneshome/atopic-dermatitis · Built with Claude — Life Sciences

~34,000 knockout signatures · 8 AD immune axes · 3 orthogonal models · 2 repurposing candidates

Every approved drug for atopic dermatitis works from outside the cell.

The type-2 allergic program is not written at the membrane. It is written by transcription factors, inside the CD4 T cell — STAT6, GATA3, and the network around them.

That is the layer nobody has screened.

OUTSIDE THE CELL · TODAY

Biologics neutralise one cytokine at a time, by injection. JAK inhibitors are oral, but blunt. ~1 in 3 patients never reach EASI-75.

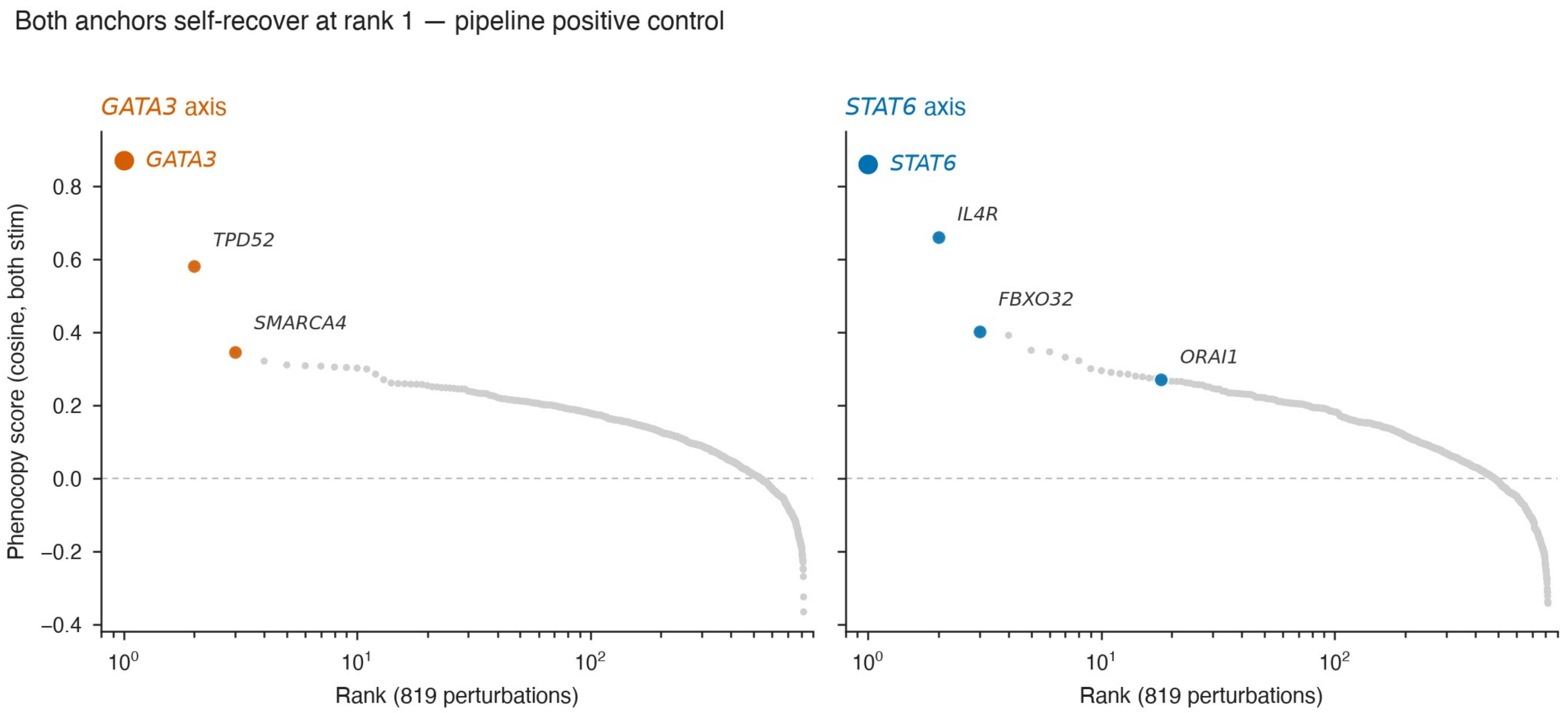
INSIDE THE CELL · THE FRONTIER

Remove a master regulator and the whole cytokine module collapses — not one output at a time. Long called undruggable, for want of an active site.

PROOF IT CAN BE DONE

STAT6 is now the first of them degraded orally (KT-621, Phase 1). Degraded, not inhibited — and a pill, not a needle.

Takeaway Which other intracellular nodes phenocopy collapse of the type-2 program — and of those, which are plausibly tractable to a degrader?

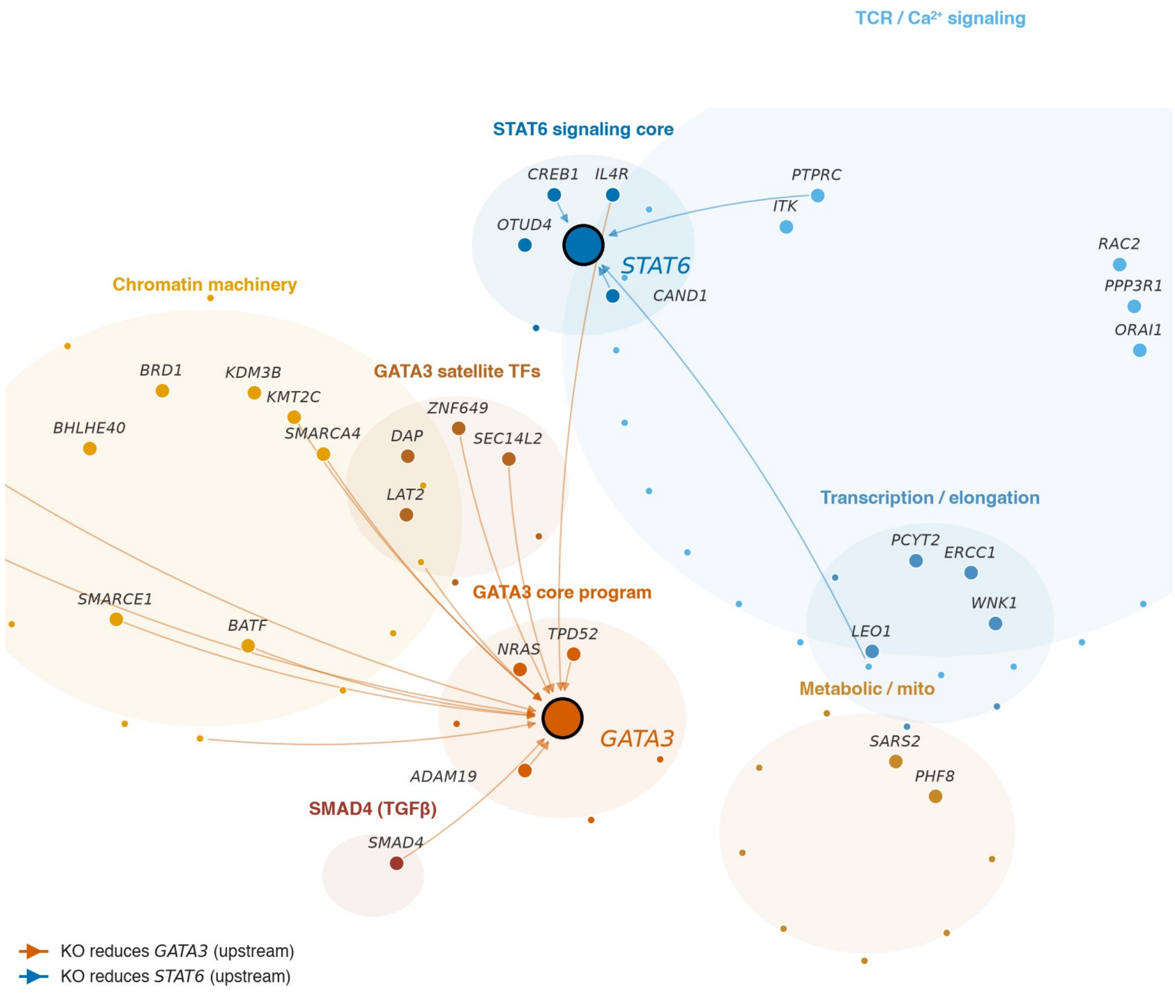


Takeaway Score every knockout for how well it phenocopies the STAT6/GATA3 collapse signature, and both anchors return at rank 1 in both conditions. The pipeline re-derives its own positive controls.

Data: genome-scale CD4⁺ T-cell Perturb-seq atlas — Zhu, Dann, Yan et al., bioRxiv 2025 (Marson & Pritchard labs). ~34,000 perturbation × condition signatures.

Intracellular Th2 network — two independent arms from perturbation signatures

A *GATA3* cytokine-program arm (warm) and a *STAT6* signaling arm (cool); the chromatin module sits upstream of *GATA3*.



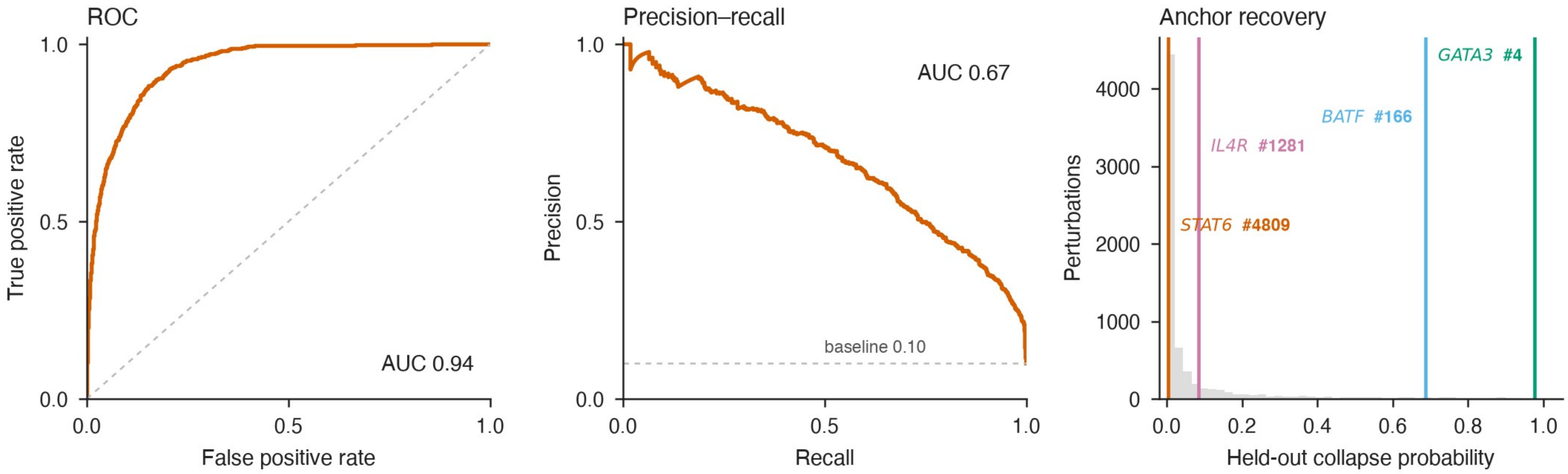
Two arms, not one

- Knocking out GATA3 collapses the type-2 cytokines — IL4, IL5, IL13.
- Knocking out STAT6 does not. Instead, Th1 genes de-repress: the two knockouts anti-correlate.
- So GATA3 maintains the type-2 program independently of STAT6, with a chromatin module upstream of it.

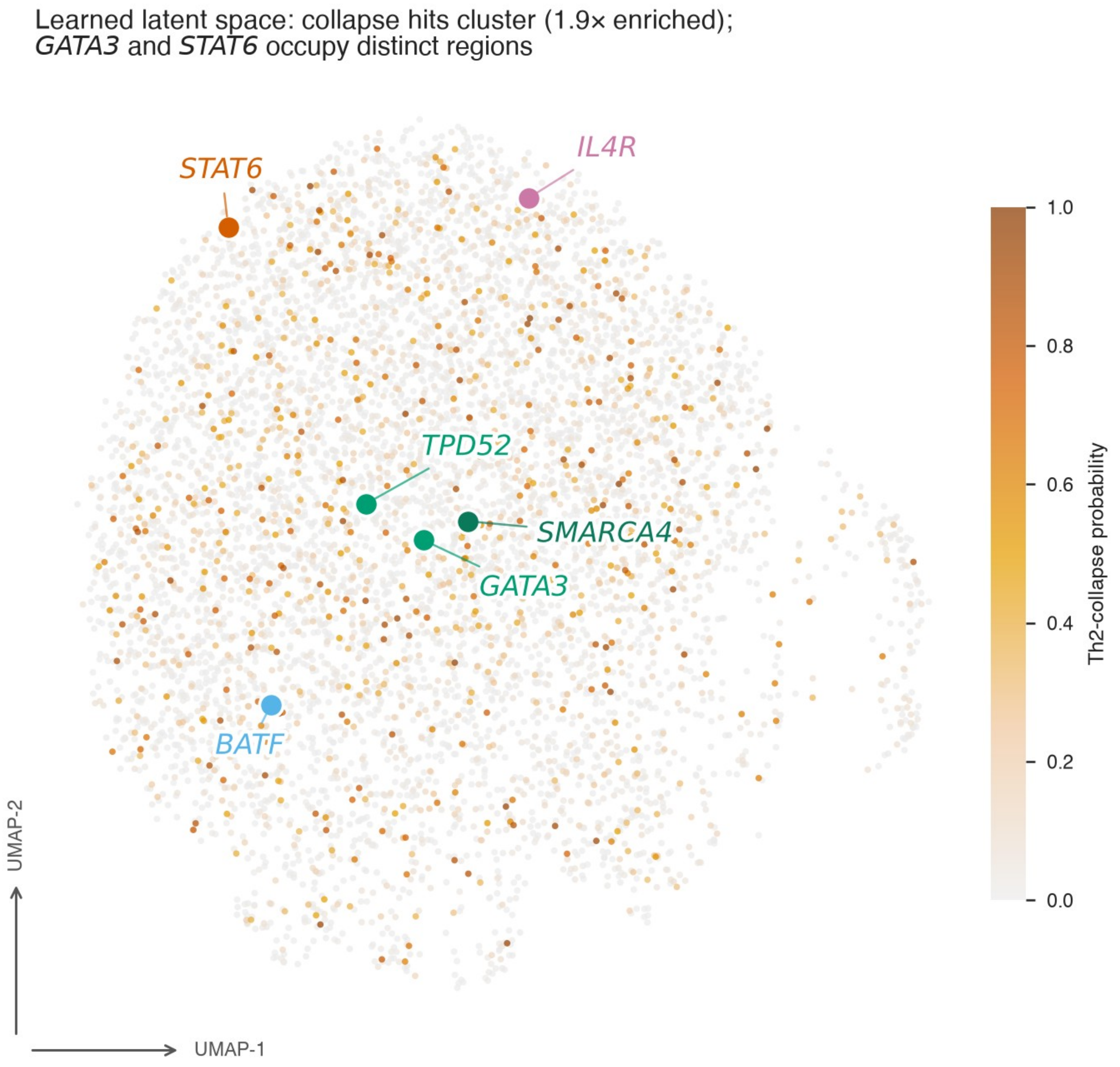
TAKEAWAY

May explain why receptor blockade fails in some patients — and predicts a STAT6-only degrader could be incomplete.

Supervised classifier recovers GATA3, not STAT6 — independent of the cosine screen



Takeaway A supervised classifier — ROC AUC 0.94, label genes held out — ranks GATA3 #4 of 6,923 perturbations, and STAT6 #4,809. The asymmetry survives a complete change of metric.



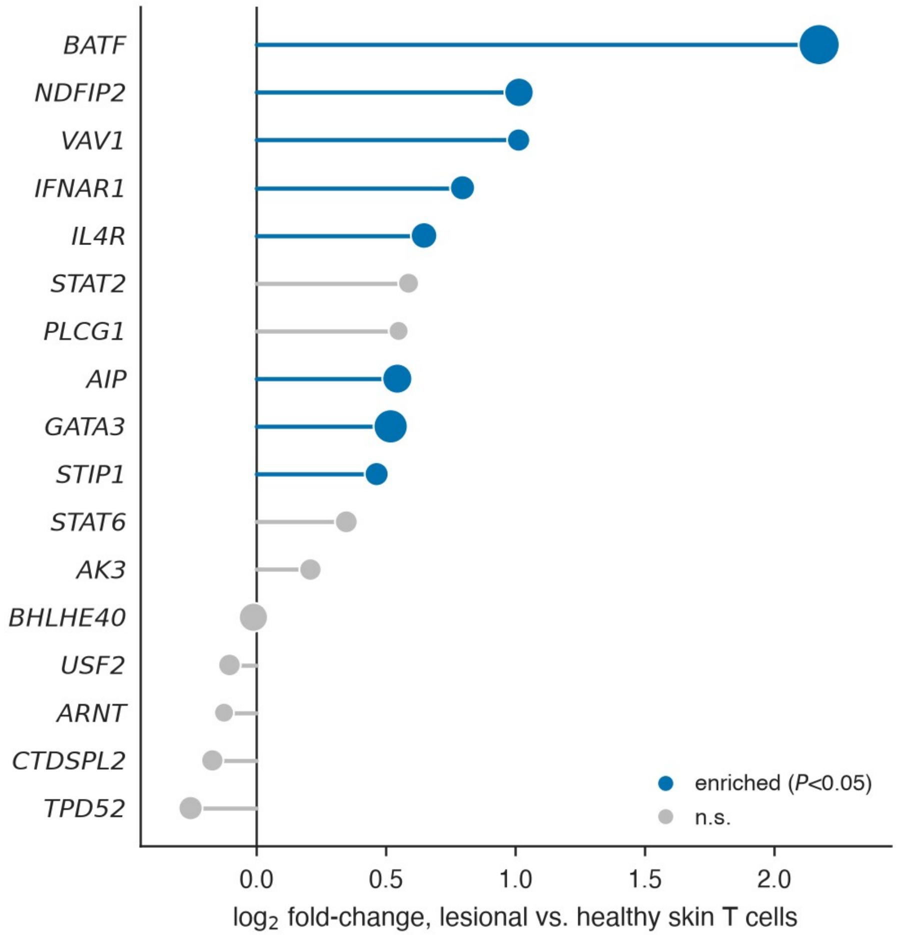
Three models, one answer

- An autoencoder, given no labels at all, puts GATA3 beside its top hit and STAT6 across the map.
- Collapse hits cluster together — 1.9× enriched.
- STAT6-independent GATA3 autoactivation is Ouyang et al., Immunity 2000; GATA3 maintenance is Yamashita et al., JBC 2004.

TAKEAWAY

The screen re-derived twenty-year-old immunology it was never shown.

Top hits are enriched in lesional AD skin T cells



Dot size ~ % of lesional T cells expressing · one-sided Mann–Whitney · GSE147424 (4 AD + 4 healthy)

Testing our own caveat

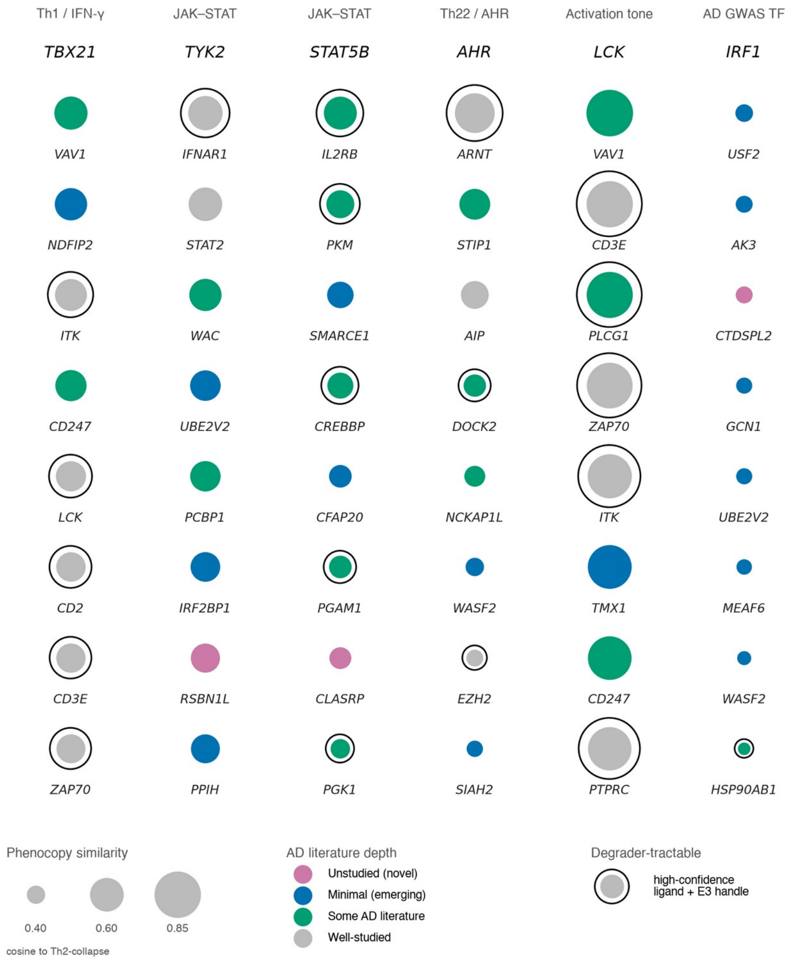
- The atlas is circulating CD4 T cells — not skin-resident Th2 cells. So we tested the hits in lesional AD skin.
- 8 of 18 hits are significantly enriched versus healthy skin — including the GATA3 anchor, BATF, IL4R and IFNAR1.
- Several hits do not transfer. The validation discriminates rather than rubber-stamps.

TAKEAWAY

The same GATA3 > STAT6 ordering reappears in tissue.

Independent validation: lesional AD skin scRNA-seq, GSE147424 (4 AD lesional + 4 healthy donors); one-sided Mann–Whitney.

Phenocopy hits across atopic-dermatitis immune axes, ranked by Th2-collapse similarity

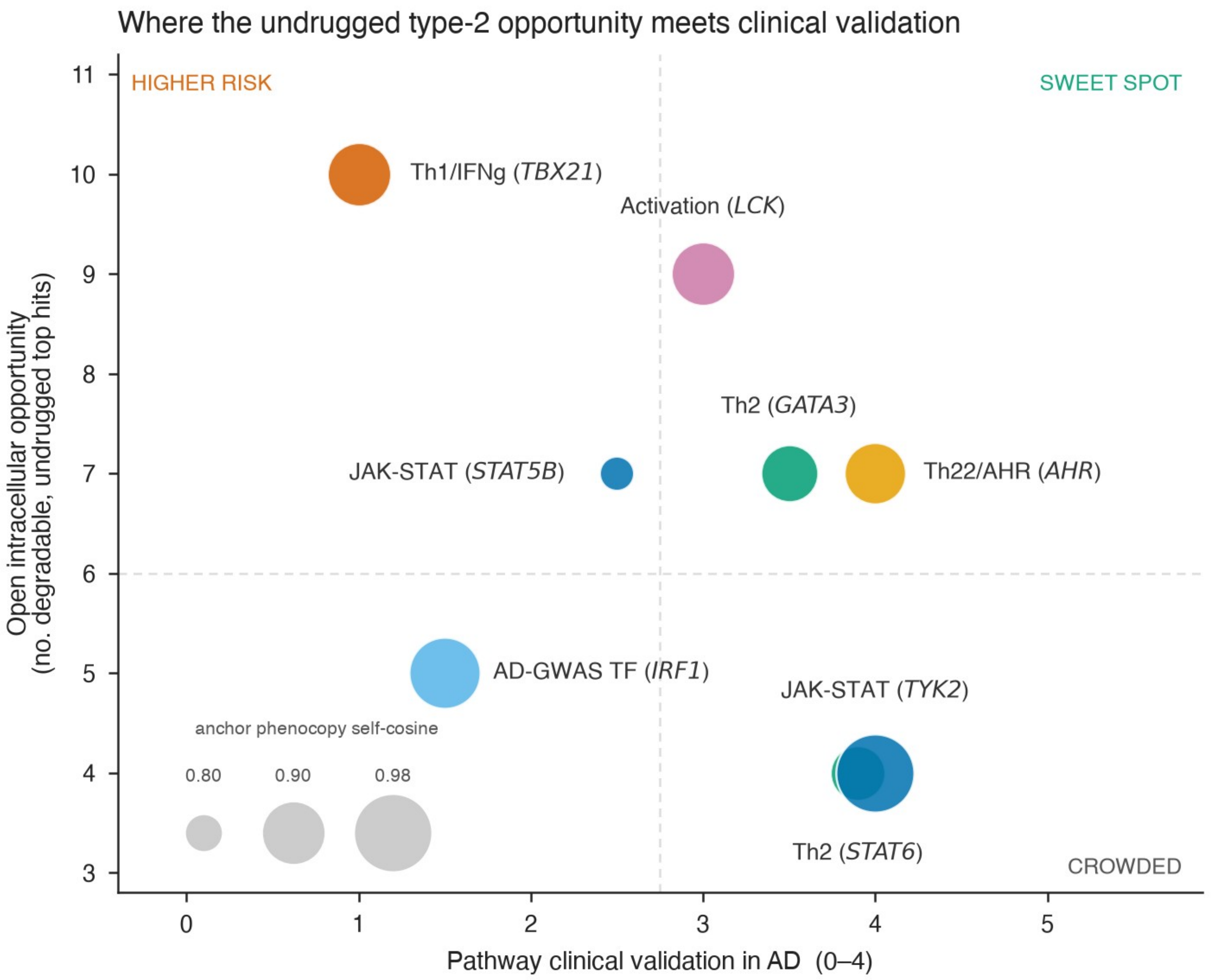


How to read it

- One column per immune axis. The anchor gene is at the top; its strongest phenocopy hits run down beneath it.
- Dot size = phenocopy similarity. Colour = how much AD literature exists (grey = well-studied → pink = unstudied). Black ring = degrader-tractable.
- Each axis re-derives its own known biology: anchor TYK2 → recover IFNAR1, its receptor. Anchor AHR → recover ARNT, its dimer partner.

TAKEAWAY

Look for ringed and pink: tractable, and nobody has studied it.

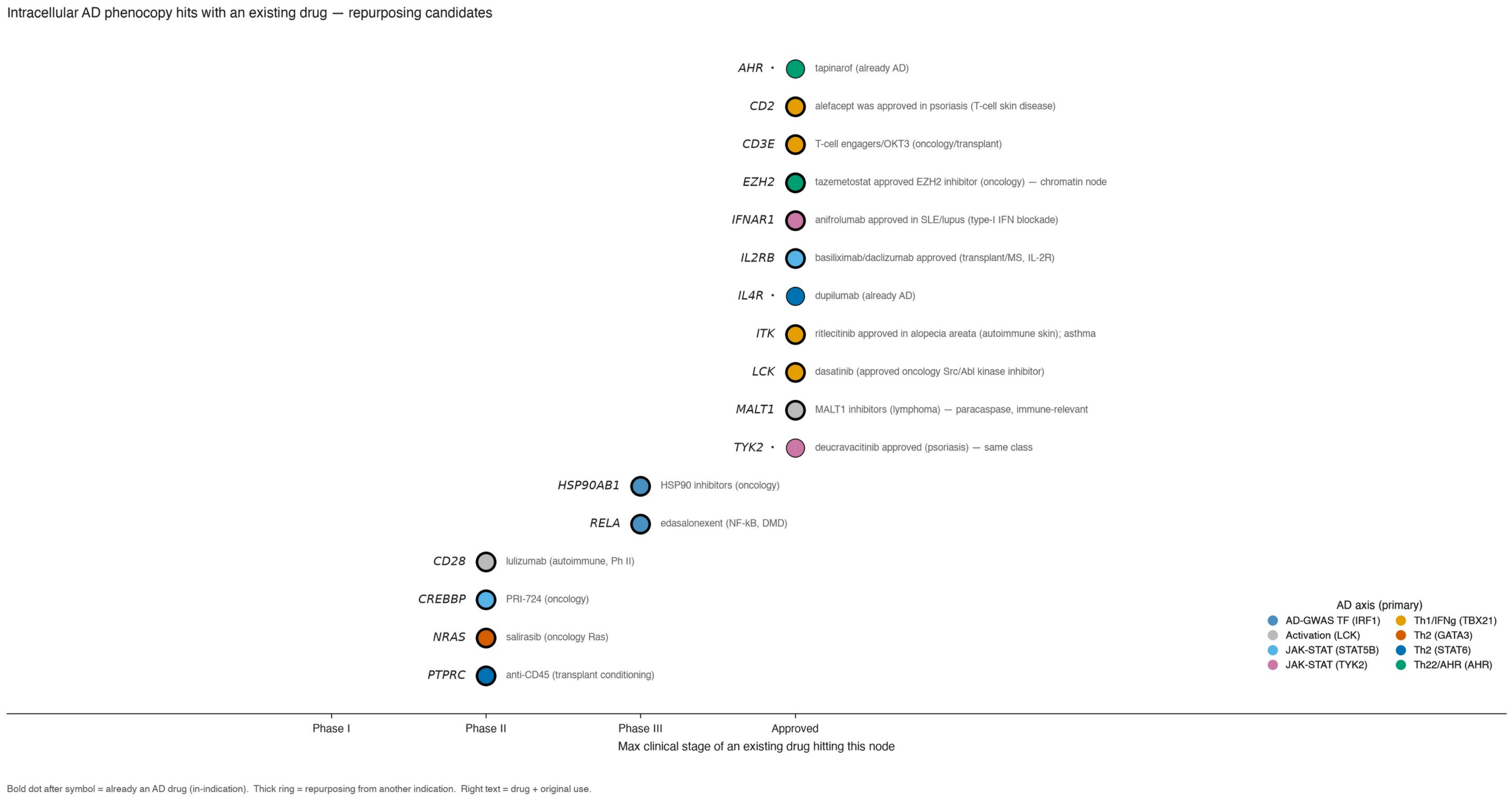


How to read it

- One bubble per axis. → right = the pathway is already clinically validated in AD. ↑ up = it still has degradable, undrugged nodes open.
- Bottom right: validated, but crowded — the field is already there. Top left: open, but no clinical validation to lean on — that is pathway risk.
- Top right is the only quadrant that is both proven and still open.

TAKEAWAY

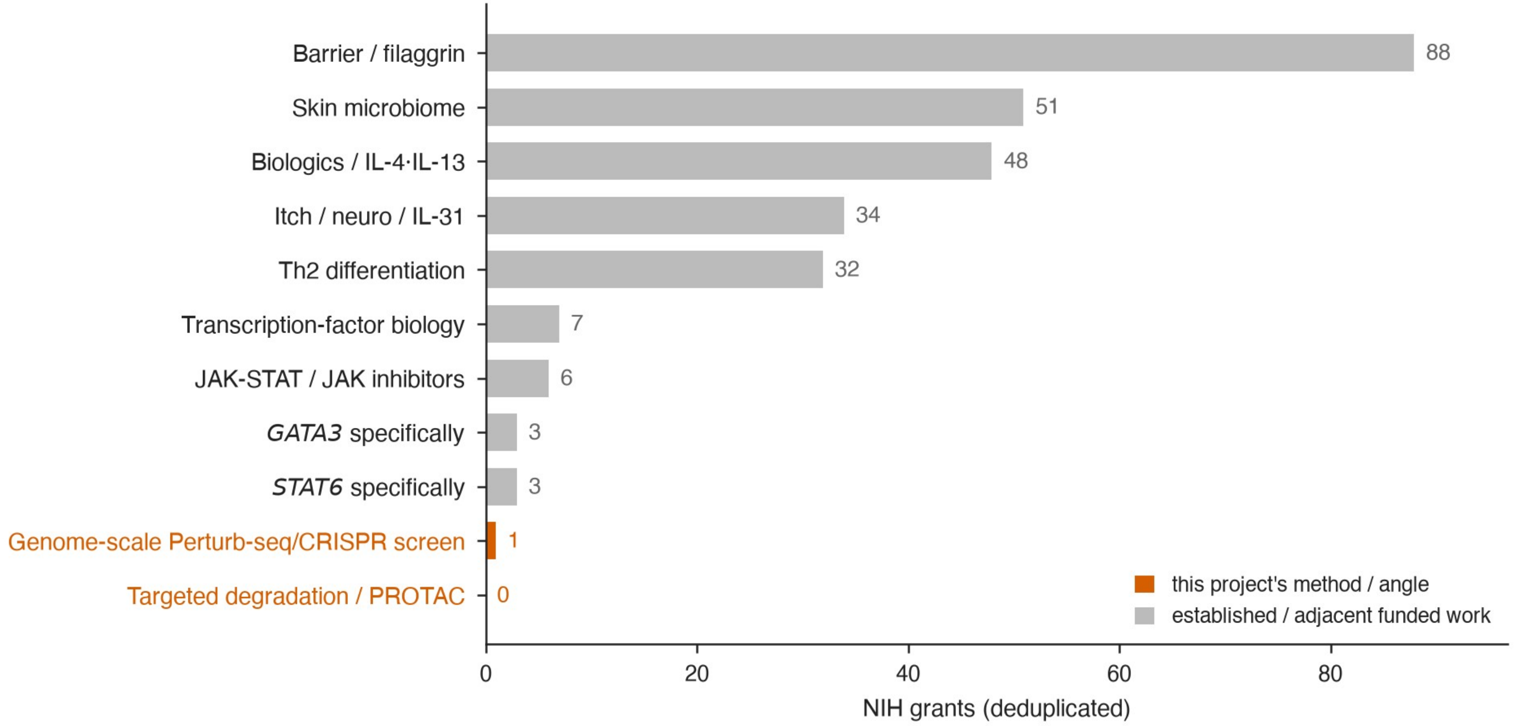
Three land there: JAK-STAT/TYK2, Th22/AHR, and type-2 itself.



Takeaway Two oral drugs, approved elsewhere, hit nodes this screen nominates and have never been tested in AD: ritlecitinib (approved in alopecia areata) and tazemetostat (an approved EZH2 inhibitor — and EZH2 sits on the chromatin module upstream of GATA3).

How to read it: one row per nominated node, placed at the furthest clinical stage any drug against it has reached. Bold dot = already an AD drug (dupilumab, tapinarof — recovered as controls). Thick ring = approved in another indication, i.e. a repurposing candidate. Colour = the axis that nominated it.

Intracellular type-2 master-regulator screening is a funding white space in atopic dermatitis



PubMed cross-check (strict queries): AD field 28,662 papers; Th2 differentiation 1,968; JAK-STAT 354; *STAT6* 101; *GATA3* 39; genome-scale Perturb-seq in AD 0; targeted degradation 1 (PMID 38262152).
Grant counts deduplicated across NIH RePORTER; AD field total n=172.

Takeaway Across 172 deduplicated NIH-funded AD projects: 3 grants on GATA3, 3 on STAT6, one genome-scale screen — and that one is a skin-microbiome study, not target discovery. Targeted degradation: zero.

NIH RePORTER, FY2022–25, deduplicated by core project number (450 raw records → 172 unique grants). All 38 abstracts in the Th2 / TF cluster read individually. PubMed queries MeSH-anchored.

What this leaves you with

1. ## A screen that validates itself

   Genome-scale Perturb-seq, three orthogonal models, both anchors recovered at rank 1 — and known biology recovered on all eight axes.

2. ## A finding that reproduces textbook immunology

   GATA3 maintains the type-2 program independently of STAT6 — confirmed in lesional AD skin, and matching Ouyang 2000 and Yamashita 2004.

3. ## Two repurposing candidates

   Ritlecitinib and tazemetostat — approved elsewhere, hitting nominated nodes, never tested in AD. In a space with zero funded degradation grants.

Sayane Shome, PhD github.com/sayaneshome/atopic-dermatitis linkedin.com/in/sayaneshome

One command re-derives the positive controls. Claude drove it all: Open Targets, GWAS Catalog, OpenAlex, ClinicalTrials.gov, GEO — every figure, and a pipeline anyone can rerun.